Fiscal Year:	FY 2004	**Task Last Updated:**	FY 01/28/2005
PI Name:	Chernov, Alexander A. Ph.D.		
Project Title:	The Role of Impurities and Convection in Macromolecular Crystal Perfections		
Division Name:	Physical Sciences		
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Program/Discipline--Element/Subdiscipline:	BIOTECHNOLOGY--Bio-nanotechnology		
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Zip Code:	35812	**Congressional District:**	5
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Task Description:	To reveal impurities influencing crystal perfection, to measure their distribution coefficients and to analyze impurity depletion zone. To check the concept of diffusion purification in microgravity. Namely, homologous impurities of which the concentration in the protein crystals typically exceeds that in mother solution (per unit volume), and are thus preferentially trapped by a growing crystal thus forming around the crystal impurity depletion zone within the solution from which the crystal grows. Growing from this depletion zone makes peripheral crystal portions more pure (at the expense of the crystal core). Therefore, major crystal volume may grow more perfect. If confirmed and developed, our impurity depletion zone concept which, we put forward in 1996-1998 brings a rational approach to biomacromolecular crystallization. This will be a step from art to science in this crystallization.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	New approach to elastic properties of crystals was put forward and implemented to successively explain ~ 10 times lower static that dynamic Young moduli. It was concluded that the static moduli are controlled by intermolecular contacts while the dynamic moduli include mainly intramolecular interactions, and intracrystalline water. The latter behaves like a solid on a nanometer scale when crystal deformation frequencies exceed ~ 107 Hz. Experiments with apo- and holoferritin were continued to evaluate distribution coefficient have been continued. Our HPLC system have been upgraded in order to measure distribution coefficients with smaller crystalline volumes with a goal to measure impurities in single crystals of a mm size scale. These preparations have been made to characterize not only the crystals grown in laboratory but also that in space. New ground based experiments have been performed to reveal role of convection on trapping of impurities. Ferritin crystals were grown in either shaken or not shaken solution samples. In the former, solution in standard wells were placed on a simultaneously inclining and rotating platform. This motion provides noticeable mixing of solution around spontaneously growing crystals. This mixing is capable of shrinking the impurity depletion zone and thus exposing the growing crystal surface to the solution containing originally introduced ferritin dimers. In the non-shaken solutions, only natural convection inducing much lower flow rates, exists. The latter influences the impurity depletion zone essentially less and thus allow purification of the mother liquid via diffusion. The crystals grown in both the shaken and not shaken wells were then harvested and dissolved in pure solvent. The resulted solution were analyzed by HPLC to determine distribution coefficient of ferritin dimers relative to the overwhelmingly present ferritin monomers – in crystal and in original solution from which the crystals grew. The average distribution coefficients were found to be 0.69 for the crystals grown in the shaken and 0.27 for the ones in the non – shaken wells. The latter lower amount of impurity trapped from a more quiescent solution supports the concept of diffusion purification which may happen in microgravity. It is important that the volumetric distribution coefficients are equal to the ones listed above times the ratio of the protein density (g/cm3) in crystal to that in solution. This ratio is of the order of 102, so that the volumetric distribution coefficient strongly exceeds unity. This is the volumetric distribution coefficients that enters diffusion equation for impurity and determines the impurity depletion zone crucial for the diffusion purification concept. Two space experiments have been prepared and implemented on board of ISS in 2004. Now the crystals were evaluated by x-ray diffraction at synchrotron and are now under investigation to measure the distribution coefficient – as compared to the samples grown in presence of convection. The work is being delayed due to the funding problems to repair HPLC system and support the workforce. This is ongoing project and the experiments are in progress. For the first time, comparison was made between distribution coefficients for ferritin crystals grown in the laboratory, either in a nearly stagnant or in a stirred solution. The latter showed more than twice larger distribution coefficient and content of impurity (ferritin dimers in ferritin crystals, acetylated lysozyme in lysozyme). These findings are in agreement with the diffusion purification concept. Ferritin crystals have been grown on the International Space Station and are now under investigation. Light scattering was applied to address the nucleation process of purified and aged catalase and of ferritin. The studies revealed strong nucleation enhancement by ageing. New approach to elastic properties of crystals was put forward and implemented to successively explain ~ 10 times lower static that dynamic Young moduli. It was concluded from calculation of liquid flow within the crysal that the static moduli are controlled by intermolecular contacts while the dynamic moduli include mainly intramolecular interactions, and intracrystalline water. The latter behaves like a solid on a nanometer scale when crystal deformation frequencies exceed ~ 107 Hz, so that the intracrystalline liquid does not have enough time to flow within deformed crystal Task Summary Series of experiments performed with ferritin, lysozyme and canavalin confirmed that homologous impurities, like dimers and other oligomers or fragments of crystallizing protein molecules are characterized by volumetric distribution coefficient essentially exceeding unity. Therefore, the impurity depletion zone around the growing crystal must exist in stagnant solution so that each next portion of the growing crystal should be purer than the previous portion. The impurity depletion zone was visualized in experiments where colorless apo ferritin crystals were grown in the presence of pink colored holoferritin dimers. Impurity depletion zone should be a reason why ~20% of crystals grown in space are of higher perfection and/or larger size. Such diffusional purification occurs at the expense of the crystal core enriched by the impurity. However, we showed that resolution of the internal stress in the core may not happen if this core does not exceed critical size. Thus, it may not compromise the crystal quality. In practice, protein purification is often insufficient, especially with respect to homologous impurities. This, we believe to be the major obstacle on the way to grow highly perfect protein crystals that are needed for structural analysis. The work is being continued to characterize the crystals grown in 2004 on the ISS in the Belgian PROMISS instrument.
Bibliography Type:	Description: (Last Updated: 01/25/2005)
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Presentation	Chernov, A.A. "Challenges in Physics of Biocrystals and Biocrystallization" Invited plenary lecture at the 10 International Conference on Crystallization of Biological Macromolecules Beijing Jun-2004